

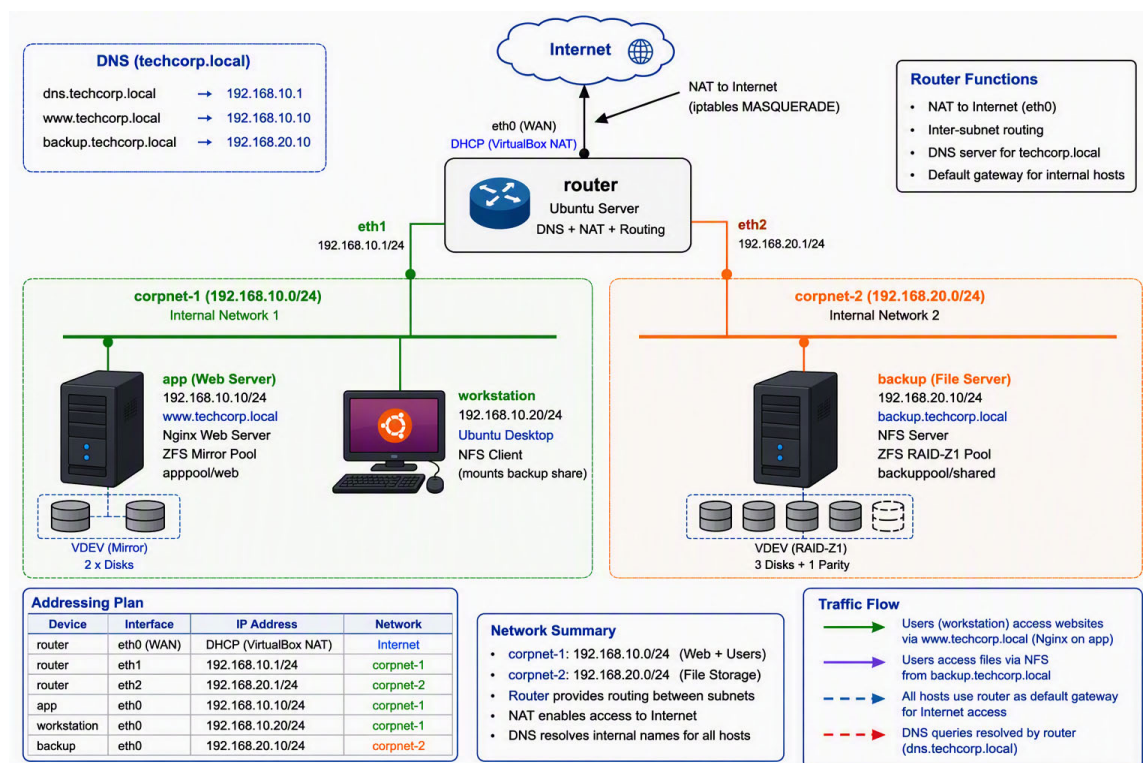
University of Wollongong

CSCI322

Assignment 3 (Due 22/11/2026)

Background

A small consulting company is building an internal infrastructure consisting of an application server and a backup server connected through two internal networks. The networks are interconnected by a Linux router that also provides Internet access and DNS services. The application server and employee workstation are connected to corpnet-1, while the backup server is connected to corpnet-2.



Tasks

- Use Ubuntu Server for the **router**, **application server** and **backup server**. Use Ubuntu Desktop for the **workstation**. Create the virtual machines in Oracle VirtualBox. Hostnames must be router, app, backup and workstation. **Use your student email ID as the username on all machines.**

- Statically configure all internal IP addresses according to the design in the diagram. The external interface of the router obtains its address automatically using DHCP from VirtualBox.
- The router must provide NAT, packet forwarding and DNS services for the internal domain techcorp.local. The systems must be reachable as www.techcorp.local, dns.techcorp.local and backup.techcorp.local.
- The application server must host an Nginx web service. The document root must be /srv/www mapped to a ZFS dataset apppool/web from a mirrored ZFS pool consisting of two virtual disks. Create a homepage displaying subject code, subject name, assignment number, student name, student number and email address.
- The backup server must host a ZFS pool backuppool created from four disks configured as RAID-Z1. Create a dataset backuppool/shared and export it through NFS.
- The workstation must automatically mount the NFS export from backup.techcorp.local at boot using /etc/fstab.

Report Requirements

- Network configuration: include screenshots of ip address output on router, app, backup and workstation. **(2 Mark)**
- Routing configuration: include the content of /etc/iptables/rules.v4 and screenshots demonstrating connectivity between networks using ping. **(2 Mark)**
- DNS configuration: include named.conf.local, forward and reverse zone files, and screenshots of dig queries for forward and reverse lookups. **(2 Mark)**
- File system configuration: include screenshots of zpool status, zfs list, showmount -e, and /etc/fstab. **(2 Mark)**
- Web service configuration: include the Nginx site configuration file and a screenshot of the web page accessed from the workstation. **(2 Mark)**

All screenshots must clearly show the username and hostname in the terminal prompt. Screenshots without these details will not receive marks.

Submission

Submit a PDF file named a3_yourUserID.pdf containing your report before the due date.